STA 137 Final Report

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Introduction

In this project, we will be analyzing the Central African Republic's economy, mainly their GDP, exports, and population from 1960 to 2017. By analyzing this data, we can further understand how their economy is going, and whether things are going in a good direction, or a bad direction. With time series analysis, we can try and see if there are specific patterns in the economy (whether it's due to advancement in technology, society, pandemics, war, famines, etc.), and prepare ourselves if there's a recession on the horizon.

Background

The Central African Republic is a landlocked country in Africa that mainly exports diamonds, gold, wood, and cash crops (Volza, 2025). The country consists of vast fertile land, water sources, and rich mines, hence the agricultural-focused economy. Due to civil unrest, corruption, and lack of reliable transportation, the country heavily relies on their exports to keep their economy going, and understanding the trends in these three variables may give us on insight on how their economy is.

Methodology

To approach this project, we will start off by plotting the data to see if there are any clear trends, seasonality, and outliers. After that, we will find their ACFs and PACFs to determine whether the data needs to be transformed and/or differenciated at varying degrees. Once their ACFs and PACFs have stabilized, we will start off with a handful of potential models such as ARMA(p,q), AR(p), and MA(q) and fit them according to what their ACFs/PACFs say.

Then, we'll perform diagnostics by plotting the residuals, finding the ACF of the residuals, plotting the QQ plot for for the residuals, and finally finding the Ljung-Box plot of the residuals. If the model does not meet the basic assumptions, then we will try to fit other more complex/simple models according to the ACF of the residuals, and perform diagnostics once again.

On top of diagnostics, we will be looking at the p-values for all coefficients, and comparing the AIC/BIC values of each potential model to help us with selecting the final model to use for forecasting.

Plotting the Data

In time-series, if there are peaks in the data, that signifies that the data is influenced by *seasonality*. If the series is not horizontal (aka it has a postive/negative trend), that means that there is a *trend* in the data, and these two factors would mean that our series *cannot be stationary*.

```
# Setup
library(ggplot2)
library(patchwork)
library(astsa)
```

```
load("Data_File_finalproject.Rdata")
data = finalPro_data

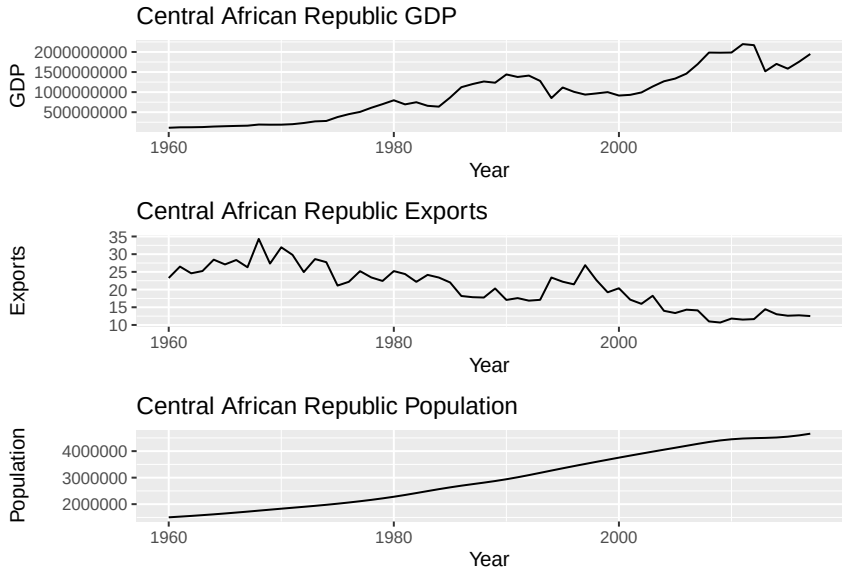
# Plots for GDP, Export, and Population
p1 = ggplot(data, aes(x = data$Year, y = data$GDP)) + geom_line() + labs(title = "Central African Republ

p2 = ggplot(data, aes(x = data$Year, y = data$Exports)) + geom_line() + labs(title = "Central African R

p3 = ggplot(data, aes(x = data$Year, y = data$Population)) + geom_line() + labs(title = "Central Africa
```

For GDP, we can see that there's a clear positive trend and peaks/dips in the data, indicating that it is **non-stationary**. For Exports, we can see a lot of sharp peaks and dips, and a clear negative trend, which also indicates that it is **non-stationary**. For Population, there are no peaks/dips, but there is a clear positive trend, which ultimately means that it too, is **non-stationary**.

Due to this, we know that we would need to difference all three variables in order to detrend and deseason our data.



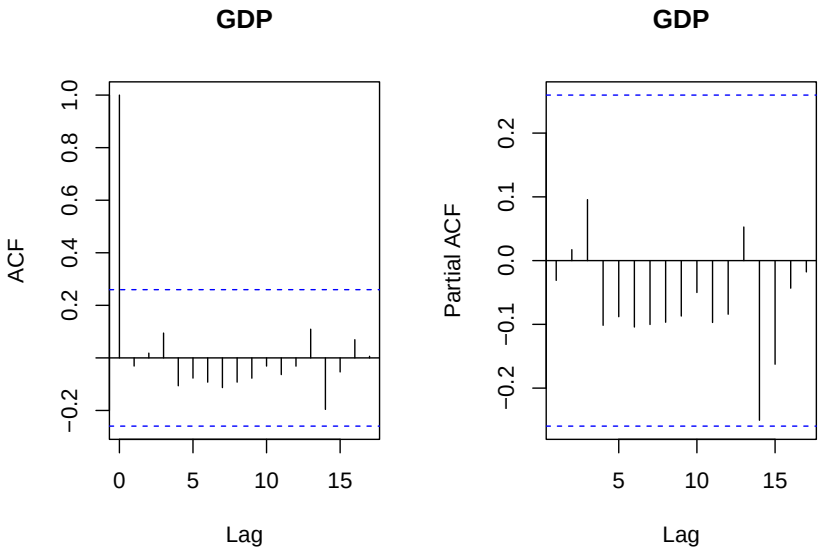
Differencing

To start off, we will perform first order differencing on our three variables, and then analyze their ACF/PACFs.

```
# Differencing for GDP, Exports, and Populations
d1 = diff(data$GDP)
d2 = diff(data$Exports)
d3 = diff(data$Population)
data2 = data.frame(Year = data$Year[-1], GDP = d1, Exports = d2, Population = d3)
```

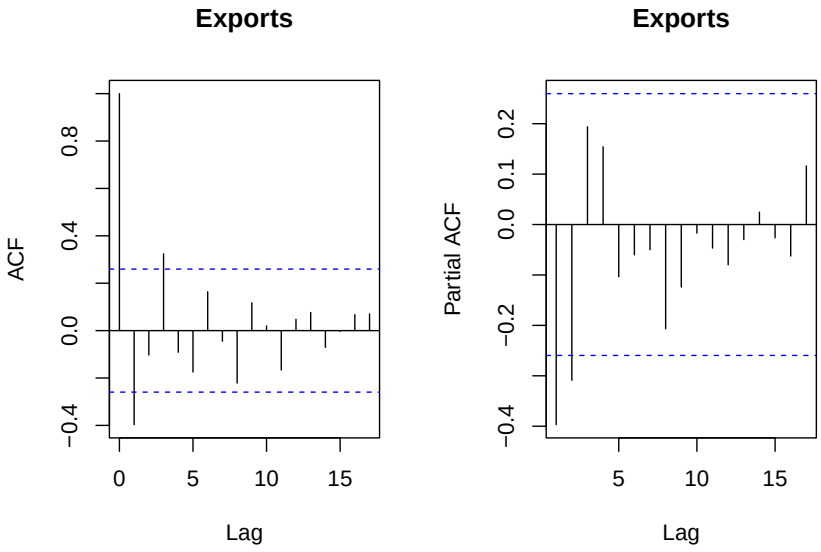
Below are the ACFs and PACFs for GDP. As we can see, all of the values beyond lag = 0 are within the blue lines, which means that the time series is **white noise**. We cannot do much with white noise, so we will attempt to perform log transformations in the next section.

```r
# GDP
par(mfrow = c(1,2))
acf(data2$GDP, main = "GDP")
pacf(data2$GDP, main = "GDP")
```



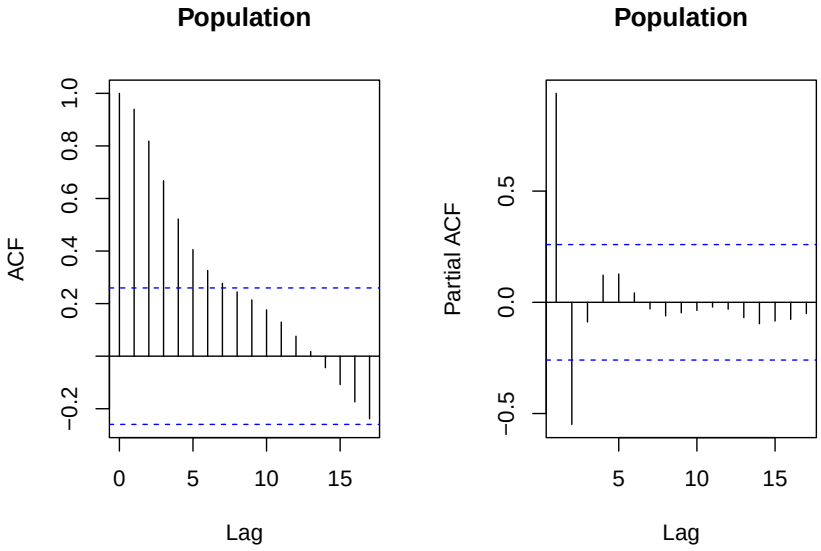
Below are the ACFs and PACFs for Exports. As we can see, both ACF and PACF decay as lag gets larger, and there is a cutoff point at lag = 3 in the ACF. This means that we can potentially model this time series with **ARMA(p,q) or MA(q)**. For now, we will attempt to perform and log transformations in the next section.

```r
# Exports
par(mfrow = c(1,2))
acf(data2$Exports, main = "Exports")
pacf(data2$Exports, main = "Exports")
```



Below are the ACFs and PACFs for Population. As we can see, the ACF looks terrible since it goes past the blue lines at lag = 7. However, the PACF appears to have a cutoff at lag = 1, and a decaying pattern, which means we could potentially fit an **AR(p) or MA(q)** model. For now, we will attempt to perform log transformations in the next section.

```
# Population
par(mfrow = c(1,2))
acf(data2$Population, main = "Population")
pacf(data2$Population, main = "Population")
```



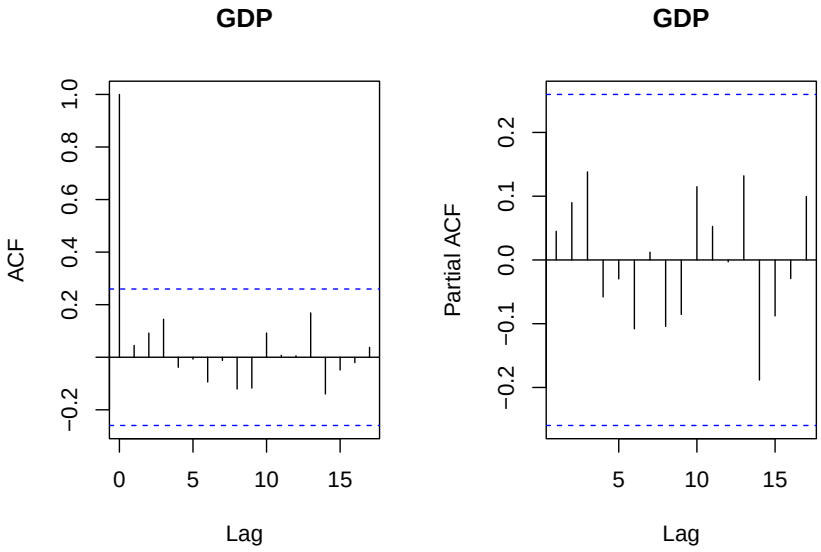
Log Transformations and Differencing at First Order

```
# Log Transformation for GDP, Exports, and Populations
log1 = log(data$GDP)
log2 = log(data$Exports)
log3 = log(data$Population)
dataLog = data.frame(Year = data$Year, GDP = log1, Exports = log2, Population = log3)

# Differencing for GDP, Exports, and Populations
d1 = diff(log1)
d2 = diff(log2)
d3 = diff(log3)
data1 = data.frame(Year = data$Year[-1], GDP = d1, Exports = d2, Population = d3)
```

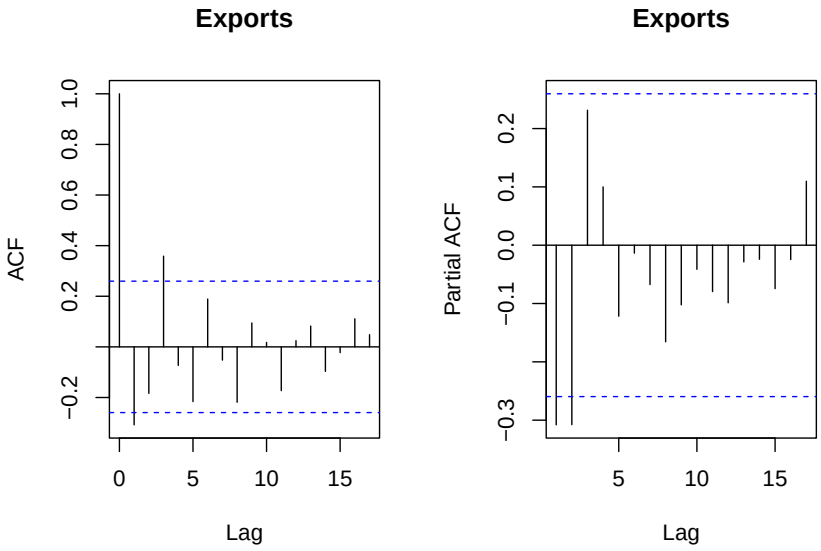
After transforming the data and differencing it at the first order, we can see that it did not make much of a difference for GDP. Both the ACF and PACF still implies that the time series is **white noise**, therefore we will experiment with second-order differencing in the next section.

```
# GDP ACF/PACF
par(mfrow = c(1,2))
acf(data1$GDP, main = "GDP")
pacf(data1$GDP, main = "GDP")
```



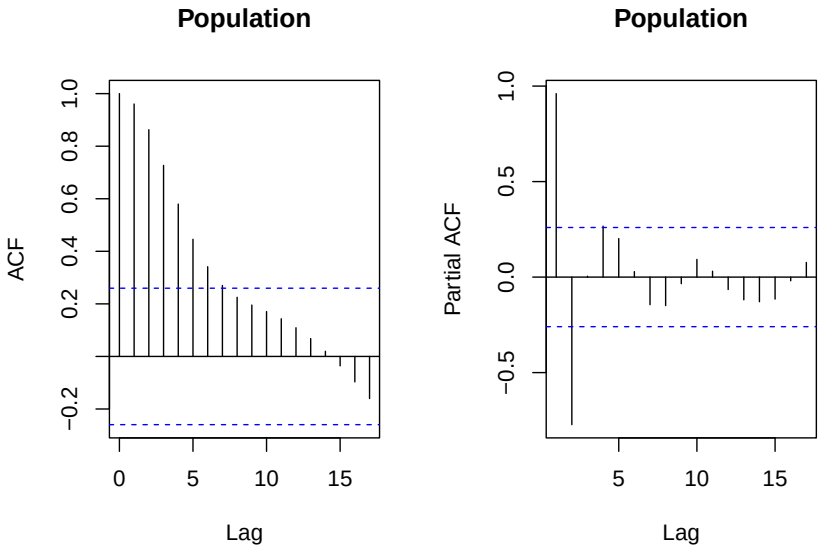
As for Exports, the ACF and PACF looks much better than before, but it looks like that we might have to fit either an **ARMA(p,q) or MA(q) model**, since the plots here have the same trends as before.

```
# Exports ACF/PACF
par(mfrow = c(1,2))
acf(data1$Exports, main = "Exports")
pacf(data1$Exports, main = "Exports")
```



For Population, the ACF is slightly better, but now the PACF is implying that we could fit an **AR(1)** model since there's a cutoff at lag = 1. In an attempt to fix the ACF, we will perform second-order differencing.

```r
# Population
par(mfrow = c(1,2))
acf(data1$Population, main = "Population")
pacf(data1$Population, main = "Population")
```

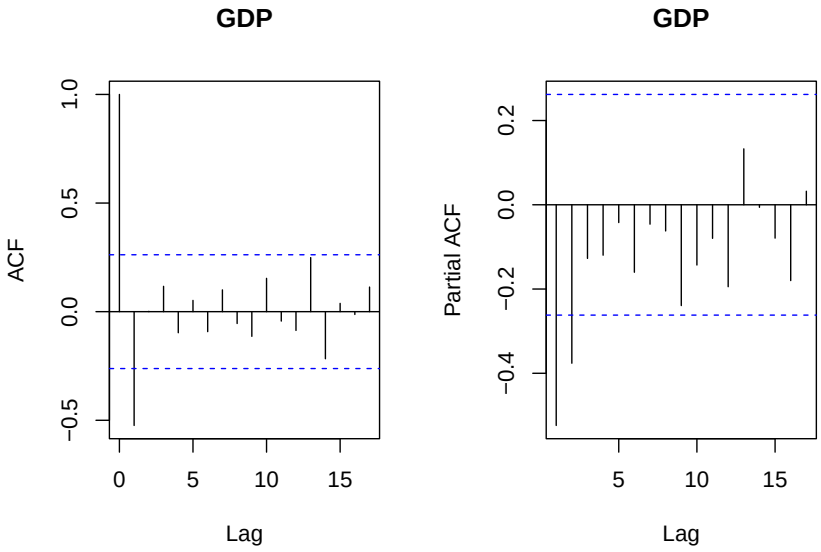


Log Transformation and Differencing at Second Order

```r
# Differencing for GDP, Exports, and Populations
d1 = diff(log1, differences = 2)
d2 = diff(log2, differences = 2)
d3 = diff(log3, differences = 2)
data2 = data.frame(Year = data$Year[-c(1,2)], GDP = d1, Exports = d2, Population = d3)
```

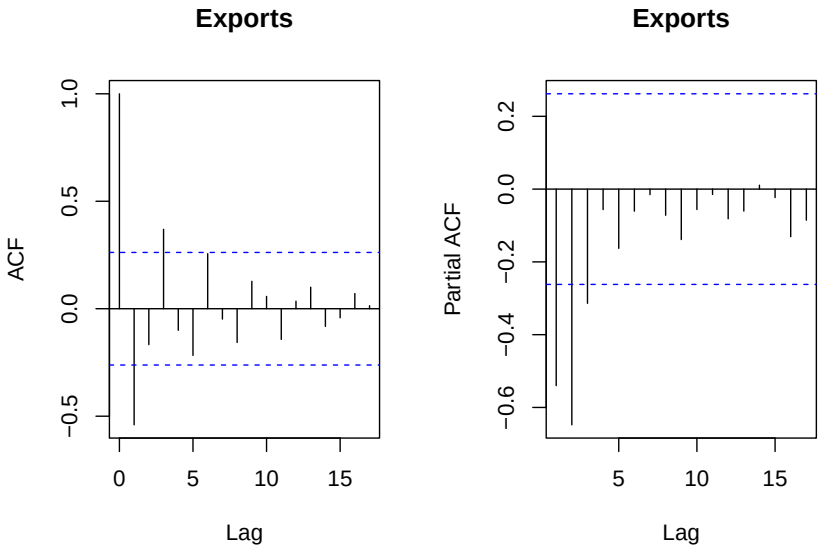
After transforming the data and performing second-order differencing, we can see that there's a cutoff in the ACF at lag = 1, and a decaying pattern in the PACF. This means that we can fit a **MA(1)** for this time series!

```r
# GDP ACF/PACF
par(mfrow = c(1,2))
acf(data2$GDP, main = "GDP")
pacf(data2$GDP, main = "GDP")
```



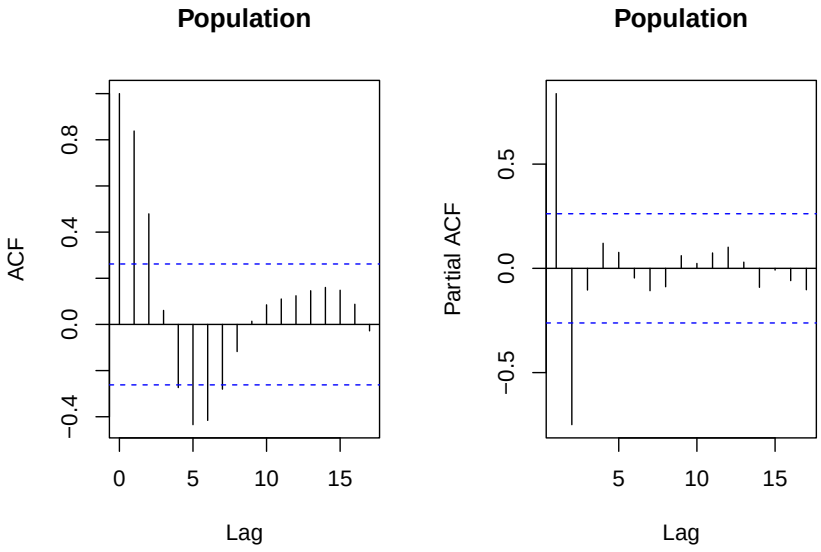
For Exports, the ACF and PACF appears to decay, and have cutoffs at lag = 3 and lag = 2 respectively. This means that we can have **ARMA(p,q), AR(2), and MA(3)** as potential models.

```
# Exports ACF/PACF
par(mfrow = c(1,2))
acf(data2$Exports, main = "Exports")
pacf(data2$Exports, main = "Exports")
```



For population, the ACF looks much better than before, but could be better. As for the PACF, it looks like it has a cutoff at lag = 1, meaning we could have **AR(1)** as our potential model. Not only that, both plots do have a bit of a decaying pattern, so we could consider **ARMA(p,q)** as well.

```r
# Population
par(mfrow = c(1,2))
acf(data2$Population, main = "Population")
pacf(data2$Population, main = "Population")
```



Model Fitting

For GDP, we will attempt to fit an **MA(1) model**, after transforming the data and differencing to the second order, and an **ARMA(1,1)** model after transforming the data and differencing to the first order.

For Exports, we will attempt to fit either an **ARMA(1,1) or MA(3) model** after transforming the data and differencing to the second order.

For Population, we will attempt to fit an **AR(1), ARMA(1,1), or ARMA(2,2) model** after transforming the data and differencing to the second order.

Summary for Each Model

According to the table below, MA(1) has smaller BIC, but ARIMA(1,1,1) has smaller AIC. For now, we will use **MA(1)** since it's more simple.

Table 1: Potential Models for GDP

	AR	AR_pval	MA	MA_pval	AIC	BIC
MA(1)	NA	NA	-0.944	0.00	-1.11	-1.038
ARIMA(1,1,1)	0.635	0.162	-0.561	0.24	-1.13	-0.991

According to the table below, we will use **ARMA(1,1)** since it has the lowest AIC/BIC values.

Table 2: Potential Models for Exports

	AR	AR_pval	MA	MA_pval	MA2	MA2_pval	MA3	MA3_pval	AIC	BIC
ARMA(1,1)	-0.297	0.024	-1.00	0	NA	NA	NA	NA	-1.16	-1.05
MA(3)	NA	NA	-1.43	0	0.546	0.172	-0.117	0.615	-1.16	-1.01

According to the table below, we will use **ARMA(2,2)** since it has the lowest AIC/BIC values.

Table 3: Potential Models for Population

	AR	AR_pval	AR2	AR2_pval	MA	MA_pval	MA2	MA2_pval	AIC	BIC
AR(1)	0.893	0	NA	NA	NA	NA	NA	NA	-11.4	-11.3
ARMA(1,1)	0.853	0	NA	NA	0.806	0	NA	NA	-12.2	-12.1
ARMA(2,2)	1.162	0	-0.444	0.001	1.544	0	0.971	0	-13.3	-13.1

Final Models

The final model for GDP will be:

$$MA(1) = -0.944W_{t-1} + W_t$$

According to the table above, the p-values for the coefficients are 0, meaning it is **very significant** to the model.

The final model for Exports will be:

$$ARMA(1,1) = -0.297X_{t-1} - W_{t-1} + W_t$$

According to the table above, the p-values for the coefficients are <0.05, meaning they are **very significant** to the model.

The final model for Population will be:

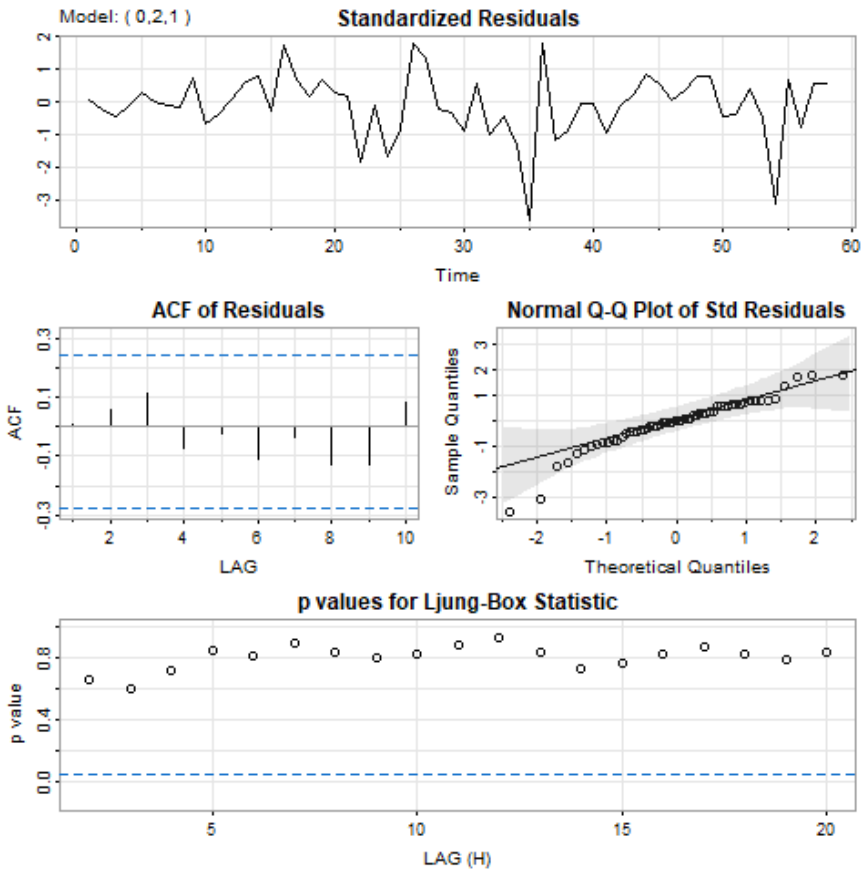
$$ARMA(2,2) = 1.161X_{t-1} - 0.444X_{t-2} + 1.544W_{t-1} + 0.971W_{t-2} + W_t$$

According to the table above, the p-values for the coefficients are <0.05, meaning they are **very significant** to the model.

Diagnostics

Ljung-Box test, QQ-Plot, ACF of Residuals, Plot of Standardized Residuals

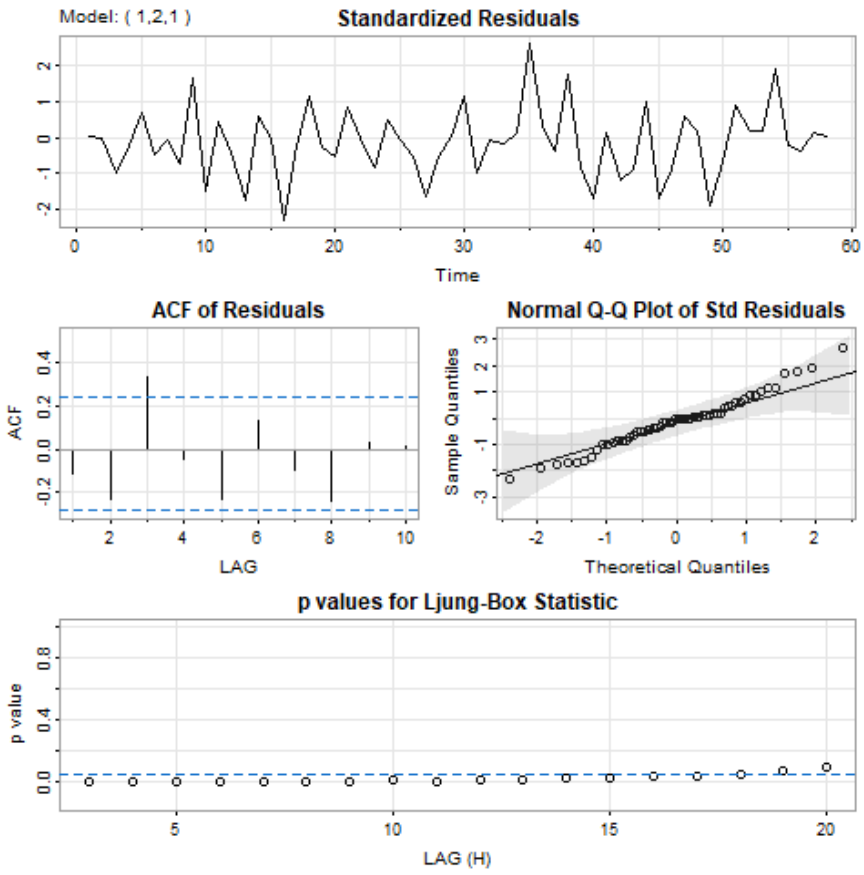
Below are the diagnostic plots for the model for GDP.



According to the Ljung-Box plot, all of the p-values are >0.05, meaning we **fail to reject** H_0, and conclude that the residuals **are independent**. As for the QQ plot, most of the residuals fall on the line, aside from two outliers. This means that the residuals are **normally distributed**. For the ACF of the residuals, they all fall within the blue lines, meaning they are **white noise**.

From here, we can confidently say that we can **proceed with the MA(1) model** for forecasting.

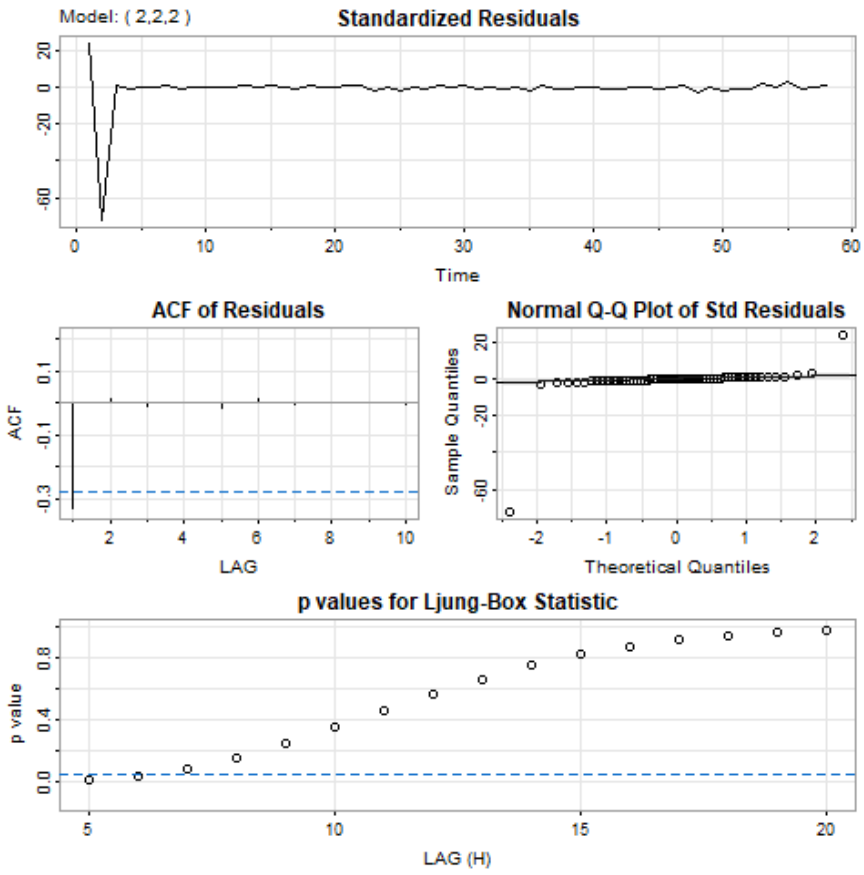
Below are the diagnostic plots for the model for Exports.



According to the Ljung-Box plot, most of the p-values are <0.05, meaning we **reject** H_0, and conclude that the residuals are **not independent**. As for the QQ plot, most of the residuals fall on the line, aside from two outliers. This means that the residuals are **normally distributed**. For the ACF of the residuals, it appears to **decay**, but most of the other values fall within the blue lines.

This means that we should try fitting a different model, such as **AR(1), AR(2), AR(3), or AR(4) with second order differencing**.

Below are the diagnostic plots for the model for Population.



According to the Ljung-Box plot, most of the p-values are >0.05, meaning we **fail to reject** H_0, and conclude that the residuals are **independent**. As for the QQ plot, most of the residuals fall on the line, aside from two outliers. This means that the residuals are **not normally distributed**. For the ACF of the residuals, it appears to have a **significant cutoff at lag = 1**, but most of the other values fall within the blue lines. Not only that, the plot of residuals show that there's a **very influential point at t = 2**, which may be skewing the results of the diagnostic plots.

This means that we can either try to remove the influential point, or fit a different model, such as **MA(1) with second order differencing**.

Fitting New Models

Below are the estimated coefficients and p-values for the coefficients for each AR(p) model. We can use AR(3) since it has smallest AIC/BIC values. Also all of the p-values for each coefficient in AR(3) are <0.05, meaning they are **significant**.

Table 4: New Potential Models for Exports

	AR	AR_pval	AR2	AR2_pval	AR3	AR3_pval	AR4	AR4_pval	AIC	BIC
AR(1)	-0.540	0	NA	NA	NA	NA	NA	NA	-0.569	-0.496
AR(2)	-0.893	0	-0.642	0	NA	NA	NA	NA	-1.082	-0.973
AR(3)	-1.108	0	-0.936	0	-0.321	0.013	NA	NA	-1.158	-1.013
AR(4)	-1.129	0	-0.993	0	-0.388	0.051	-0.06	0.656	-1.125	-0.945

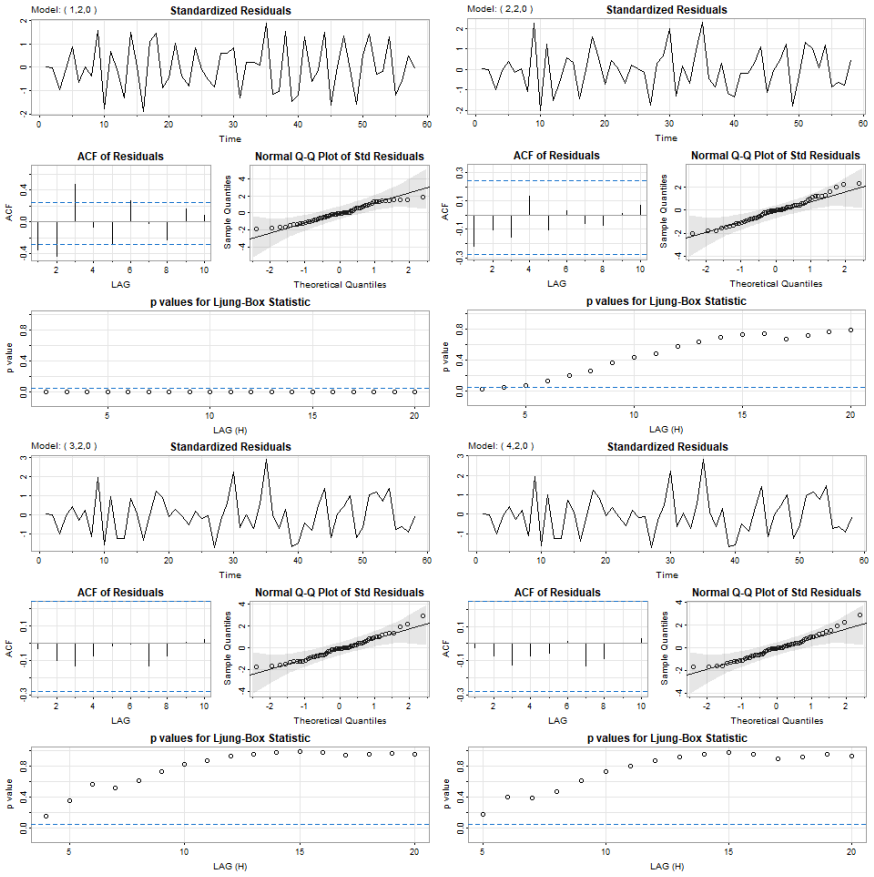
Below are the estimated coefficient and p-values for the coefficient. Since the p-value is <0.1, we can say that it's **significant**.

Table 5: New Model for Population

	MA	MA_pval	AIC	BIC
MA(1)	0.999	0	-11.1	-11.1

Diagnostics on New Models

Below are the new diagnostic plots for the AR(1), AR(2), AR(3), and AR(4) models for Exports.

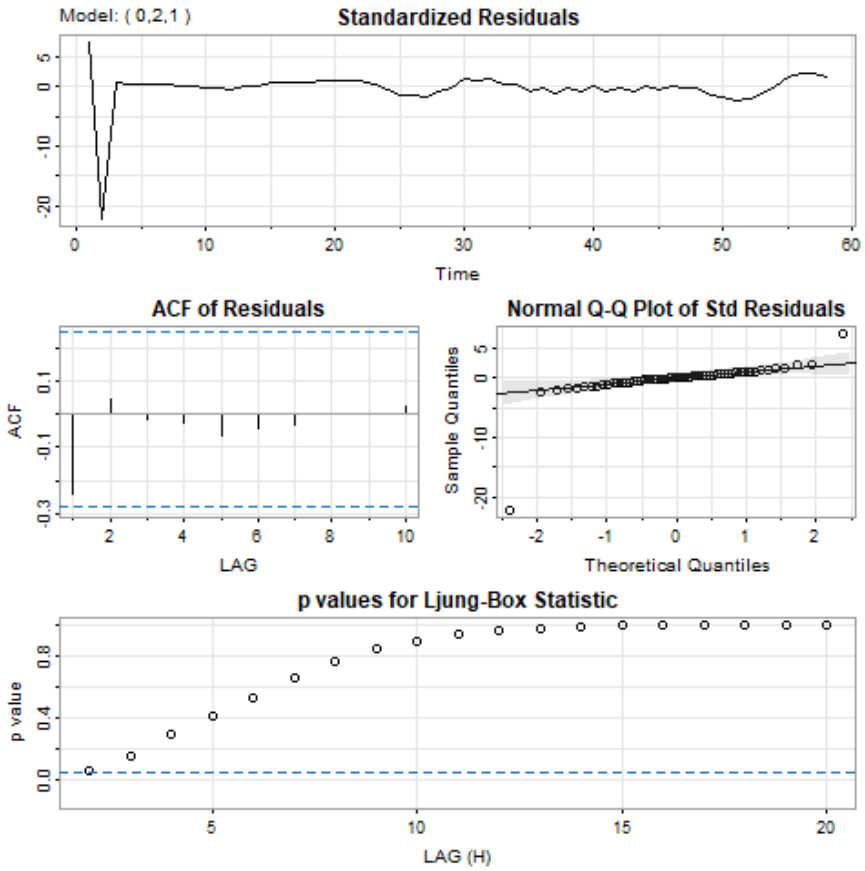


As we can see, all of the plots of the residual ACFs are within the blue lines for AR(2), AR(3), and AR(4). This means that the residuals for those models are **white noise**. The QQ plot for those three models also show that the residuals mostly fall on the line, meaning they are **normally distributed**. Finally, the Ljung-Box plots for those three models show that most of the p-values are >0.05, meaning the residuals are **independent**. This means that these three models **meet all basic assumptions** required to qualify as a good fitting model.

According to the Ljung-Box plots, the p-values get better and better as we incorporate more coefficients, however these improvements cap out at AR(3). The Ljung-Box plots for AR(4) appear to perform slightly worser than AR(3), while the other plots (QQ plot, ACF of residuals, plots of residuals) remain mostly the same between the two. These results also line up with the values that we got from the table above.

Due to this, we can confidently **proceed with AR(3) as our final model** for forecasting.

Below are the new diagnostic plots for the MA(1) model for Population.



According to the Ljung-Box plot, most of the p-values are >0.05, meaning we **fail to reject** H_0, and conclude that the residuals **are independent**. As for the QQ plot, most of the residuals fall on the line, aside from two outliers. This means that the residuals are **normally distributed**. For the ACF of the residuals, they all fall within the blue lines, meaning they are **white noise**. Unfortunately, the plot of residuals still shows that there's a **very influential point at t = 2**.

Despite that, we will still **proceed with the MA(1) model** for forecasting.

Forecasting

To summarize, the models that we will be using for each variable are:

- GDP: $MA(1) = -0.944W_{t-1} + W_t$, where σ^2 is 0.0173.
- Exports: $AR(3) = -1.108X_{t-1} - 0.936X_{t-2} - 0.321X_{t-3} + W_t$, where σ^2 is 0.0155.
- Populations: $MA(1) = 1.77W_{t-1} + W_t$, where σ^2 is 0.000000735.

Below are the forecasting plots for GDP, Exports, and Population:

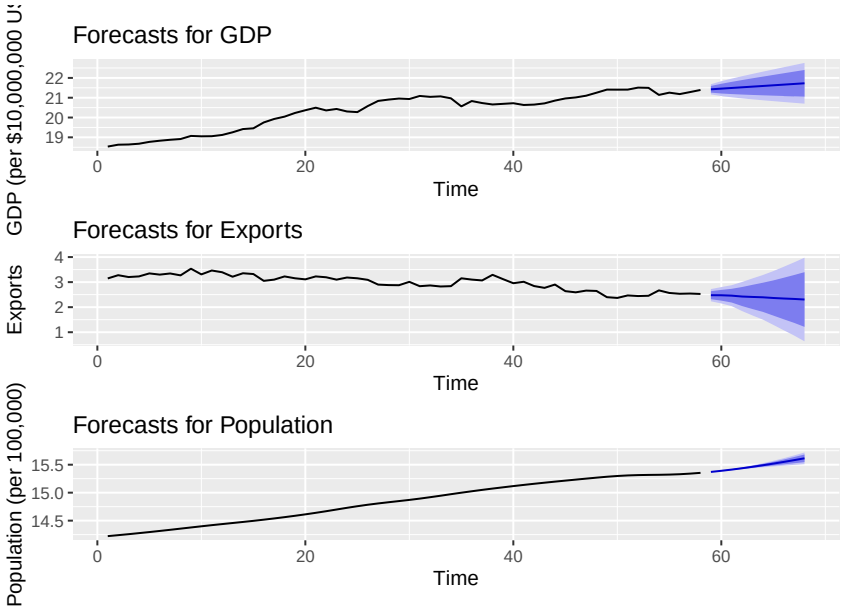
```r
library(forecast)

# Forecast for GDP
GDParima = arima(dataLog$GDP, order = c(0, 2, 1))
GDPforecast = forecast(GDParima, level = c(80, 95))
f1 = autoplot(GDPforecast) + labs(title = "Forecasts for GDP", y = "GDP (per $10,000,000 USD)")

# Forecast for Exports
Exportsarima = arima(dataLog$Exports, order = c(3, 2, 0))
Exportsforecast = forecast(Exportsarima, level = c(80, 95))
f2 = autoplot(Exportsforecast) + labs(title = "Forecasts for Exports", y = "Exports")

# Forecast for Population
Poparima = arima(dataLog$Population, order = c(1, 2, 0))
Popforecast = forecast(Poparima, level = c(80, 95))
f3 = autoplot(Popforecast) + labs(title = "Forecasts for Population", y = "Population (per 100,000)")

f1 / f2 / f3
```



According to our forecasts, the GDP is estimated to very slightly increase over the years, with some fluctuations. For exports, it's estimated to very slightly decrease over the years, although it may or may not fluctuate rapidly. For the population, it's estimated to increase at a steady rate.

Conclusion

During our exploratory analysis, we were able to find that none of the variables were stationary, and all three of them had a trend. However, population was the only variable that didn't have seasonality. While performing model selection, we found that all three variables required transformed data and second-order differencing just to stabilize the ACF/PACF, and be able to draw some sort of conclusion.

During diagnostics, the model that we initially chose for the GDP was perfect (MA(1)), but the models for exports and population needed some reworking. After going through the model selection process again, and

performing diagnostics on those new potential models, we were finally able to settle with a decent model for both variables (AR(3) and MA(1)).

Once we were able to do some forecasting with the models, both GDP and exports didn't have any sort of strong trend, but they did have a rather wide range of potential values. As for the population, the trend was very obvious, and the potential range of values was very narrow.

With these results, we are able to determine that the data that we were given isn't the best, due to the amount of manipulation it required just for results. Aside from that, we now have a grasp on the trends in the Central African Republic's economy. Being able to forecast potential trends allows us to prepare ourselves for the future, and have an idea of where to invest in order to improve the current trend, or to increase the probability of the trend remaining positive.

References

- Top Exports in Central African Republic in 2025
- Economy of the Central African Republic